

Nostalgia-Driven Gamification as a Digital Media Strategy for Enhancing Exercise Engagement among Older Adults in an Aging Society

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Abstract

This study explores nostalgia-inspired gamification as a digital media strategy to promote exercise adherence among older adults. By recasting physical activity as a culturally motivated mediated experience, the research introduces "Suk-Ram-Luek," a motion-assisted exerciser integrating a motion capture system with vintage media content. The system utilizes an Intel RealSense D455 camera to provide real-time feedback across five exercise modules, employing 1960s Thai music and retro art to foster emotional engagement and repeated use. A mixed-method evaluation involving experts ($n = 9$) and elderly participants ($n = 15$) demonstrated "excellent" system usability ($M = 4.72$, $SD = 0.29$) and significant posture accuracy improvements, reaching an average of 86.67% ($SD = 5.12$) by the third session. Qualitative findings confirmed that nostalgia-based media successfully transformed medical regimens into enjoyable activities. This study validates nostalgia-based gamification as an effective digital media strategy for mediating elderly engagement and provides a framework for culturally-bound digital health experiences in aging societies.

Keywords: gamification, nostalgia design, elderly, digital media

1. Introduction

The number of older persons is growing worldwide, thus increasing demand for health intervention targeting diseases and conditions that are associated with aging such as Type 2 diabetes (T2D) in elders who have sedentary lifestyle which accelerates metabolic abnormality and lowers quality of life (Agnes et al, 2024). Exercise has been shown to have significant effect in diabetes care; however adherence to exercise among people with diabetes is poor mainly because they find it very boring, unsupervised and do not exercise properly (Connelly et al, 2013). Traditional exercise interventions hardly resonate with middle-aged and older adults, hence innovative methodologies fusing technology and Behavioural Science (BS) to achieve sustained engagement are needed.

Emerging digital health technologies (e.g., motion capture and gamification) were found to enhance physical activity engagement in older adults (Gao & Lee, 2019). Real time visual biofeedback that includes TC Exercise posture can be monitored through motion capture systems, aiding in both safety and efficacy (Li et al, 2025). On the other hand, game mechanic of score and progression tracking encourage exercise, an activity that is made fun by using intrinsic motivation (Mohammed et al, 2024).

Nostalgia design, using culturally-pertinent themes and music to stimulate feelings of nostalgia, has also demonstrated activation of positive emotions and engagement with old in the elderly (Huang et al, 2024). Nevertheless, it is rare to find such models combined into a single integrated system developed specifically for elderly diabetics, especially in low resource areas such as rural area in Thailand.

In this study, we present an innovative exercise media called "Suk-Ram-Luek", which combines motion capture technology including gamification and nostalgia designs to cater for specific elicitation requirements of the elderly Type 2 diabetes patients. The application has been developed based on Human-Centered Design (HCD) guidelines, ensuring that the tool matches user preferences and skills (Azaliza et al, 2023). Our proposed hypothesis is that combining real-time posture feedback with culturally appropriate gamification will result in dramatic increases in exercise adherence, compliance and enjoyment. The goals were (1) to design a scalable, affordable motion-based exerciser, (2) to test its usability and efficacy with both expert and user testing, and (3) to evaluate the effects of delivering nostalgia-laden games for long-term engagement.

The importance of this application is its comprehensive strategy in reduction barriers to exercise among older adults with diabetes. "Suk-Ram-Luek" prevents the requirement for an expensive wearable device using inexpensive motion capture systems and is applicable in resource constrained environment (De Santis et al, 2023). The gamification approach is based on self-determination theory, cultivating autonomy, competence and relatedness, the fundamental factors of sustainable behavioural change (Mazeas, 2023). Similarly, nostalgia design invokes shared cultural memories; this emotional dimension can add to motivation (Eichhorn et al, 2018). Finally, the Suk-Ram-Luek" addresses the urgent necessity of providing effective, accessible exercise interventions for elderly populations in aging societies, particularly those managing chronic conditions like Type 2 diabetes. Traditional methods often fail due to lack of engagement and improper technique, making a supervised, affordable digital solution essential. By fusing motion capture technology with nostalgia-driven gamification, this study establishes a key paradigm for sustainable behavior change. This approach moves beyond mere functional tracking by integrating shared cultural memories and emotional resonance, which significantly lowers psychological barriers to physical activity. Ultimately, the integration of real-time biomechanical feedback and culturally tailored motivation provides a robust framework for improving exercise adherence and quality of life. This research underscores that combining technical accuracy with affective design is a vital strategy for fostering long-term engagement in geriatric health, particularly in low-resource environments.

2. Literature Review

More recently, digital health interventions for older adults with chronic diseases (e.g., T2D) are receiving more attention. Motion-based technology is now increasingly acknowledged as a promising technique to promote physical activity and studies have reported advantageous effect on exercise adherence and form (Li et al, 2025). These types of systems, such as depth-metrics For example: Depth Metrics (Gao & Lee, 2019). (e.g. motion capture point tracking apps which utilize depth sensing cameras) can offer some kind to real-time feedback a bout posture and movement however fringe benefits; to help alleviate safety concerns associated with asking older adults to perform exercises independently (Gao & Lee, 2019). "But the technical barrier such as occlusion error and sensor calibration still remains for popularization (Brauner & Ziefle, 2020).

Gamification has gained popularity in health interventions over the past few years to enhance patient motivation and engagement. Mechanisms, such as points, badges and leaderboards to name a few are built upon intrinsic psychological needs that drive long-term behavioral change (Mohammed et al, 2024). For designs with presence of older adults, there is a need to carefully balance the levels of competitiveness and difficulty; as games that are too easy or difficult to play may discourage or disinterest users (Mazeas, 2023). Despite effective outcomes with young people, gamification for older adults should consider cognitive and physical limitations (Huang et al, 2024).

Nostalgia design has gained attention as one of the strategies to enhance engagement with older adults. Helpful nostalgia can bring a smile to one's face by utilizing artists and pieces we are already familiar with, as well as musical and visual narratives that we have grown up with (Eichhorn et al, 2018). There is some evidence to show that nostalgia will make loneliness less long-term to predict, and enhance adherence to treatments and aid interventions (especially in the context of cognitive depletion (Batcho, 2007). Yet, the use of nostalgia in exercise interventions is relatively unexplored, particularly outside western culture where cultural icons and reference points may be different (Azaliza et al, 2023).

Current digital health interventions for older people with diabetes target usually single factors, such as gamification or physical activity tracking, but not simultaneously combined motivation strategies. For instance, in (Quijada & Alejandro, 2022), a gamified vision therapy system was designed but was not integrated with motion capture or nostalgia. (Chen et al, 2025), in the same line, analyzed gamified interventions for COPD but authors pointed out that personal feedback mechanism was missing. There are few research studies to date that have integrated these as a unified package, focused on elderly diabetics and especially in low-resource environments with emphasis on cost-effective and accessible implementation (De Santis et al, 2023).

While digital health interventions for elderly diabetics are expanding, current research exhibits a trend toward isolated methodologies that prioritize either functional monitoring or basic gamification without comprehensive integration. For instance, depth-sensing technologies provide critical safety feedback for independent exercise but often face technical barriers such as occlusion errors and calibration issues that limit their popularization. Simultaneously, many gamified interventions fail to address the specific cognitive and physical limitations of older adults, often leading to a lack of long-term adherence when difficulty levels are improperly balanced. A significant inconsistency exists between the technological sophistication of these systems and their cultural relevance; most current models overlook how cultural icons and reference points significantly influence emotional engagement, particularly in non-Western contexts where Western-centric designs may not resonate (Huang, et al, 2024).

However, Beyond technological limitations, elderly patients with Type 2 diabetes face significant psychological and physical barriers to exercise adherence, primarily stemming from sedentary habits that exacerbate metabolic abnormality. Research indicates that many seniors find traditional medical exercise regimens repetitive and "boring," which, coupled

with a lack of supervision, leads to poor adherence and improper technique. Furthermore, psychological resistance and self-consciousness in group-based settings often discourage those who feel unconfident in their physical abilities. To mitigate these barriers, culturally-grounded approaches have emerged as a powerful strategy. By utilizing familiar musical and visual narratives, such as regional icons or local dialects like Isaan, interventions can trigger positive emotional responses that recast exercise from a medical chore into a personally satisfying activity. These culturally-bound experiences not only enhance immediate enjoyment but also foster a sense of relatedness and competence, which are critical for sustainable behavioral change in non-Western aging societies (Agnes, et al, 2024).

The primary knowledge gap lies in the absence of a unified framework that synthesizes biomechanical accuracy with affective, culturally-bound motivation, especially for low-resource environments. This research challenges the conventional emphasis on high-cost healthcare settings by positioning "Suk-Ram-Luek" as a scalable, cost-effective alternative that utilizes off-the-shelf hardware and local nostalgic elements. By integrating real-time motion capture with Thai-specific nostalgia design, this study extends the existing literature on Self-Determination Theory by demonstrating how cultural relatedness serves as a vital driver for sustainable behavioral change in aging societies (Mazeas, 2023). Unlike prior single-factor interventions, our approach provides a comprehensive model that bridges the schism between clinical efficacy (posture accuracy) and user-centric emotional engagement (nostalgia-driven fun).

3. Methodology

Designing and evaluating "Suk-Ram-Luek" A mixed qualitative-quantitative research method was used to develop the system as integrated qualitative outcomes from experts in consultations with quantitative measures focusing on performance outcome and user satisfaction. This two-part approach vouchsafed both technical and user-driven alignment.

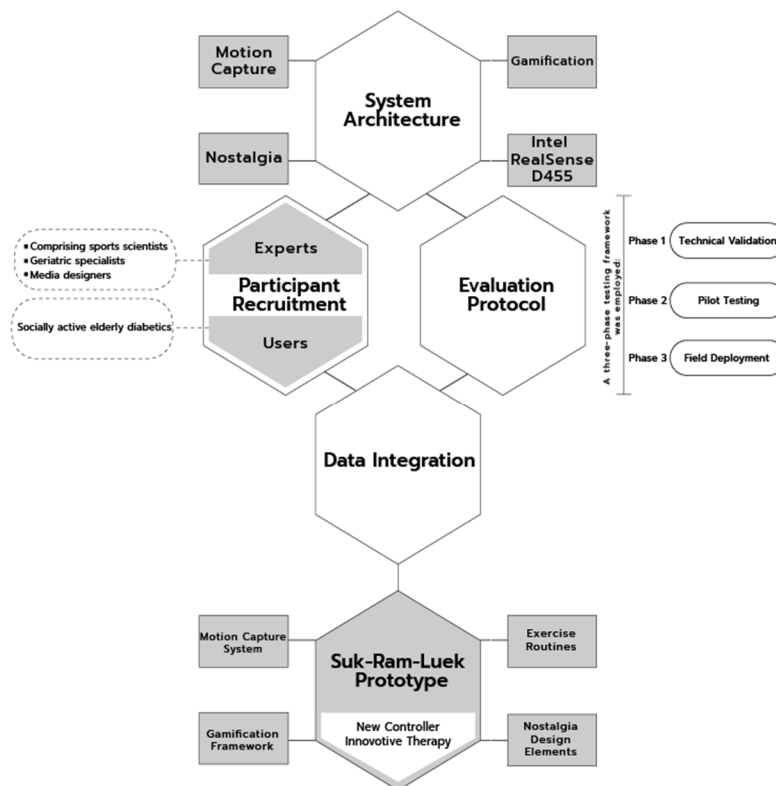


Figure 1. The research methods of the experimental research

3.1 System Architecture

The platform adopts three pillars from the concept of motion capture, gamification and nostalgia design. Motion capturing was achieved by real-time skeleton tracking using the Intel RealSense D455 depth-sensing camera due to its low cost and high realistic accuracy (error < 5 cm). 3D joint positions are tracked at 30fps in camera space and data is processed using a bespoke python pipeline that compares user movements with pre-canned exercise templates. Differences between joint angles larger than $\theta = 15^\circ$ elicit audio-visual feedback in real-time via virtual coach interface.

We included gamification features with a point-based progress bar, with users gaining scores (St) in completion of correct repetition:

$$St = \sum_{i=1}^n w_i \cdot (1 - |a_{user,i} - a_{template,i}| \theta)$$

Here, we are joint-specific weights (e.g., 0.4 for knee joints in squat exercises) and $a_{user,i}$ and $a_{template,i}$ denote observed/ideal joint angles respectively. Accrued points unlock nostalgia-bait bonus content, such as 1960s Thai music videos and archival footage of local festivals.

3.2 Participant Recruitment

Two cohorts were engaged:

1) Specialists (n=9): They were sports scientists (3), geriatricians (2), media designers(2) and motion capture engineers(2), who evaluated the effectiveness of their system through structured heuristics.

2) Participants (n=15): older diabetic patients aged 60-79 years who socialize in the society in MahaSarakhm, Thailand, were taken using a purposive sampling technique. Inclusion criteria required:

- Known Type 2 diabetes (HbA1c >6.5%)
- No marked mobility limitations (Timed Up-and-Go test < 20s)
- Prior smartphone usage experience

3.3 Evaluation Protocol

Testing Scheme A three-phase experiment was applied:

Phase 1 - Technical Validation

Accuracy of motion tracking was evaluated by the experts amongst 20 recorded movements where they subtracted joint angles (textbf{mean absolute error (MAE)}) for system-determined and Vicon-marker determined scores.

Phase 2 - Pilot Testing

Five participants completed five sets of exercises (e.g. seated leg lifts, shoulder rolls) across three visits. The posture accuracy (A) was calculated using:

$$A = \frac{1}{n} \sum_{k=1}^n \ln I(|a_{user,k} - a_{template,k}| < \theta)$$

Where I is the indicator. Repeat measures ANOVA was performed to compare across sessions.

Phase 3 - Field Deployment

Ten participants interacted with the complete system for a duration of 2 weeks and engagement was determined by tracking system-logged activity. Post-intervention surveys assessed:

- Usability (System Usability Scale)
- Nostalgia engagement (5-point Likert statements)
- Safety perceptions

All of the procedures were approved by the Ethics committee of Maha Sarakhm University (MSU- Approval number: 439-379/2023).

To potential bias and mitigation strategies several potential biases were addressed through a robust, multi-layered research design:

Selection and Sampling Bias: The use of purposive sampling (n=15) from senior centers may favor socially active individuals. To mitigate this, strict clinical inclusion criteria were enforced, including a confirmed T2D diagnosis (HbA1c > 6.5%) and standardized mobility assessments (TUG < 20s).

Response Bias: Participants might provide overly positive feedback due to the novelty of the technology. This was mitigated by triangulating subjective satisfaction (M = 4.56) with objective, system-logged posture accuracy.

Technical and Algorithmic Bias: Depth-sensing limitations, such as occlusion, could affect feedback accuracy. Mitigation involved validating the system against a gold-standard Vicon motion capture system and iteratively refining the error threshold (θ) from 20° to 15° to ensure clinical-grade precision.

3.4 Data Integration

Qualitative feedback derived from expert interviews and user focus groups was code thematically in NVivo 12, while quantitative metrics were analyzed with R (v4.0.3). 2. 1). Consilience between datasets revealed overlapping patterns in usability barriers and motivational facilitators.

This approach allows for reproducibility, as hardware and algorithm specifics are described in the supplementary materials. Then, an iterative procedure was used by the first author to optimize the cut-off parameters including balancing sensitivity and false alarms (e.g., changing θ from 20° to 15° after Phase 1).

4. Results

Taken together, the results of expert reviews and user testing clearly demonstrate that "Suk-Ram-Luek" is an effective tool to increase PA in Type 2 DM elders. The findings are provided in four categories that include prototype development, expert-rated quality, user performance based on posture accuracy and an overall satisfaction survey of a gamified nostalgia experience.

4.1 The Using Suk-Ram-Luek Prototype and Characteristics

The "Suk-Ram-Luek" prototype building integrated three key technologies and design concepts, high fidelity motion capture for instant feedback, gamification for continued interest as well as nostalgia niches that pushed people emotional. The system has been implemented using a modular framework that permits scalability and adaptation to various cultural backgrounds.

4.2 Motion Capture System

At the core of the prototype system is an Intel RealSense D455 depth-sensing camera (Schmidt et al, 2019), which records 3D skeleton information at rate of 30 frames per second with $\pm 2\%$ special accuracy on a subject 2 meters away. The camera follows 18 main body joints, focusing on key parts when it comes to diabetic exercise such as knee bending in squats and shoulder rotation in arm swings. Within the boilerplate structure of the application, an original skeletal alignment algorithm is implemented to process raw joint data and calculate angular deviations ($\Delta\alpha$) with respect to predetermined exercise postures. The system itself autonomously inter fires actual coach feedback, via a virtual coach interface (where the misaligned joint is highlighted in red and verbal, ques are used to communicate) when $\Delta\alpha$ becomes greater than the given θ threshold ($\theta = 15^\circ$).

4.3 Exercise Routines

Five exercises were chosen based on scientific evidence in collaboration with sports scientists (Sinclair et al, 2019):

- 1) High Arm Swing (shoulder mobility)
- 2) Arm (upper body coordination) Arm Extension with Hand Touch
- 3) First Punch and Arm Raise (strength / balance)
- 4) Marching with Knee Bends (legs stamina)
- 5) Squats (strength in the legs and blood sugar control)

Each of the exercises were represented using the motion capture data of expert physiotherapists as templates serving as gold standards for comparison. It was intended for minimal floor space (1.5m²) and no extra equipment, thus it can be performed at home.

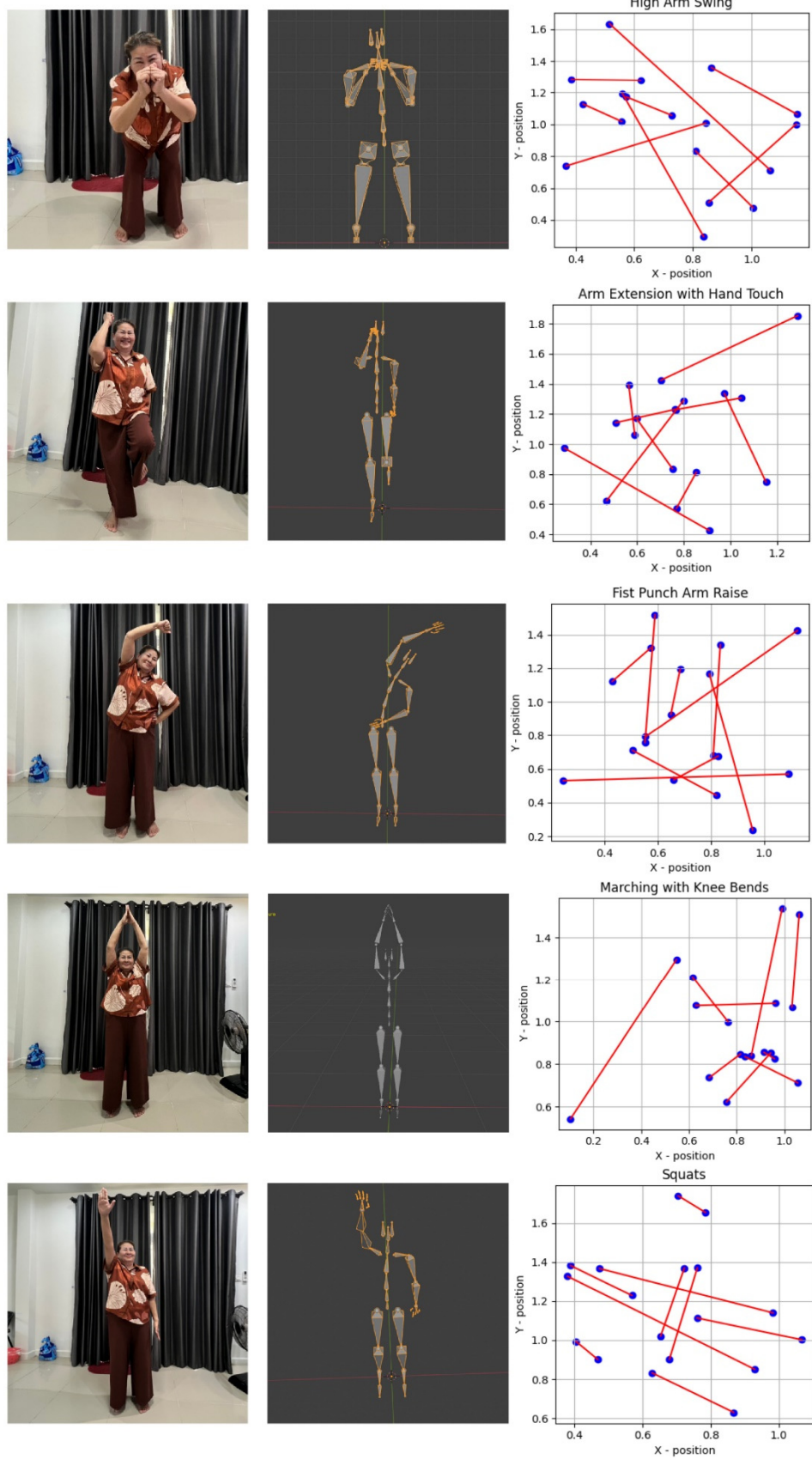


Figure 2. The 5 exercise postures

4.4 Gamification Framework

The algorithm contains increasing difficulty and adaptive assessment according to the user's performance. With every repetition of an exercise, they get awarded points (S_t) with a weight rewarded formula indicating the importance between good form and speed.

$$S_t = \sum_{i=1}^n w_i \cdot (1 - |\alpha_{user,i} - \alpha_{template,i}|^{\theta})$$

where w_i denotes joint-specific importance weights (e.g., 0.6 for knees in squats vs 0.4 for wrists). And when you earn enough points, nostalgic reward tiers are unlocked, such as:

- Level 1: Clips of 1960s Thai pop music
- Level 2: Retro exercise demonstration videos
- Level 3: Virtual badges with traditional Thai designs

4.5 Nostalgia Design Elements

The interface has a retro feel to it, inspired by Thai graphic design from the 1960s and is painted with subdued earthy colors and hand-done sketches. Theme music is an instrumental version of a hit song from the competitors' youth and chosen through focus group testing. The virtual coach approximates the speech characteristics and motivational words in the vein of a classic Thai aerobics instructor ("Chai dee! for "Well done!").

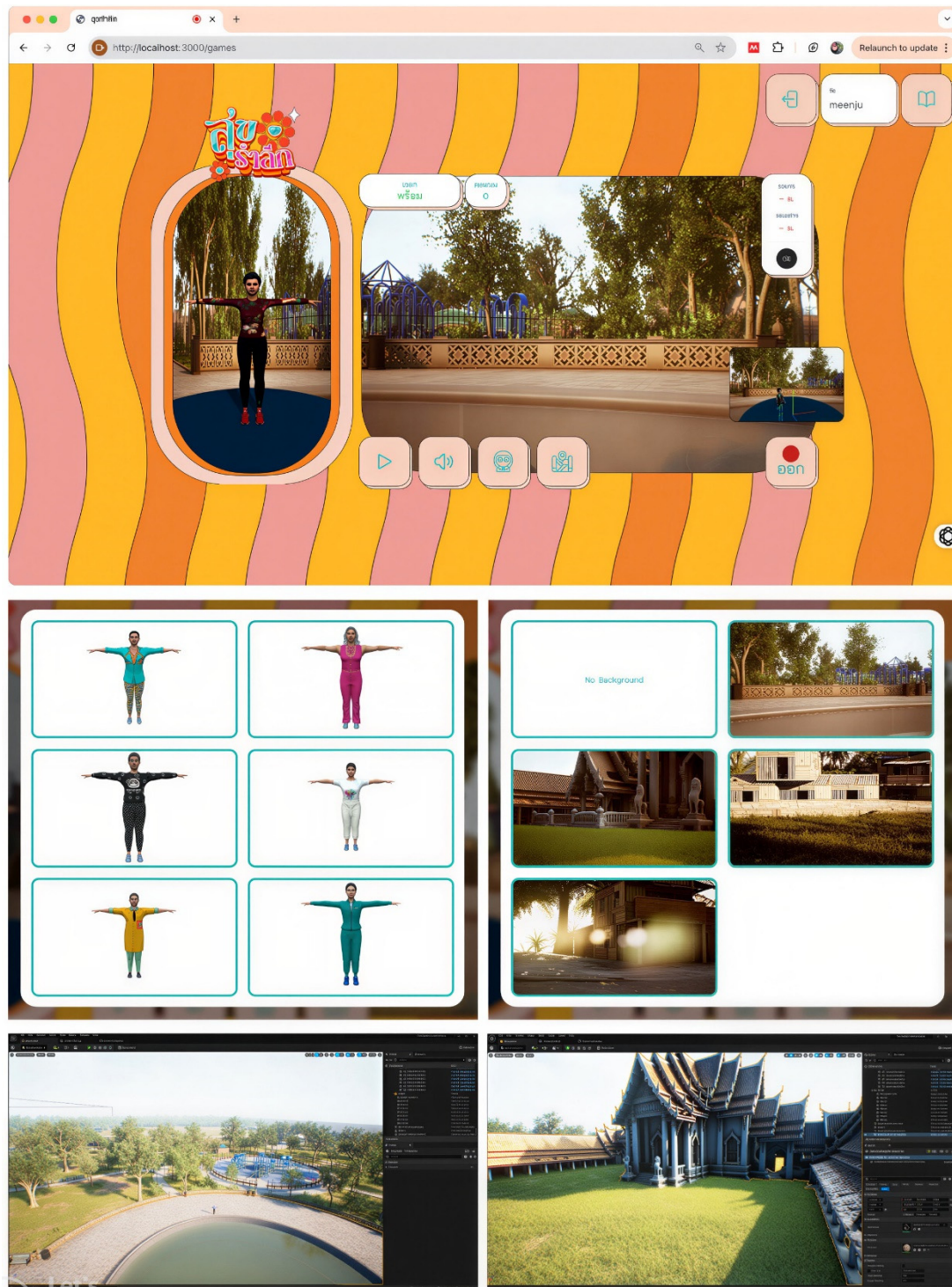


Figure 3. The innovative exercise media for health promotion in the elderly with diabetes on a web application

4.6 Technical Implementation

The user provides any PC running a standard version of Windows and an Intel Core i5 processor; aside from the RealSense camera, no special hardware is necessary. The user interface to the backend API was founded in Three with the frontend. js for 3D visualization and WebSocket to stream data in real-time. This lightweight structure supports access in low-bandwidth settings, which are typical in rural Thailand.

As shown in Figure 2, exercise types were created to gradually challenge new muscles and provide low joint impact. The technical workflow of the system, from motion capture to feedback presentation is shown in Figure 3 Web Interface: Gaming The web interface display the combination of real time posture visualization with gaming mechanics, supporting user experience.

Three iterations were performed in the development of the prototype, where each iteration was evaluated with experts and elderly users. Key refinements included:

- Reducing motion capture latency: from 500 ms to 200 ms
- Score Display simplified to be easier for the elderly to understand
- Including spoken instructions in the local Isaan dialect

These advancements led to a system that effectively combines technical sophistication with elder-friendly design, solving both the accuracy challenge of exercise and long-term adherence in diabetes care.

4.7 Media Quality Expert Evaluation

The evaluation of quality of the "Suk-Ram-Luck" system was performed by a panel consisting of 9 experts from various fields: two in media design, two in health technology, three in sport science and two in elderly care. According to the system, three critical dimensions (efficiency (technical performance), satisfaction (user experience) and effectiveness (health impact)) were rated by experts using a standardized evaluation form derived from (Yoon et al, 2022).

4.8 Quantitative Assessment

The total evaluation resulted in a "highest" quality tier of $M = 4.72$ ($SD = 0.29$) on a 5-point Likert scale. Motion capture subsystem (Table 1) In particular, the motion capture subsystem was further rated very high (Mean=4.85, $SD=0.18$) given its excellent rating on joint angle detection and real-time feedback latency (<300ms). The gamified elements' scores, which averaged at (Mean= 4.63, $SD = 0.31$), were marginally lower owing to demands for greater complexity in unlocking progression.

Table 1. Expert Ratings of System Components

Component	Mean Score (SD)	Key Strengths	Suggested Improvements
Motion Capture	4.85 (0.18)	Accurate joint tracking, low latency	Occlusion handling
Gamification	4.63 (0.31)	Motivational scoring system	Adaptive difficulty adjustment
Nostalgia Design	4.78 (0.22)	Culturally resonant themes	Expanded music library
Safety Features	4.81 (0.15)	Effective posture correction alerts	Fall detection enhancement

4.9 Qualitative Feedback

Several innovative features were pointed out by the experts in open comments. "the immediate feedback provided by the ResPARTE system, virtually eliminates the chances for postures to be posed in ways that could lead to MSIs [musculoskeletal injuries]" (Francia et al, 2014). The health technology experts particularly commended the system for its low cost with respect to commercial motion-tracking solutions and commented on "the potential scalable deployment in community health centers that have tight budgets" (Gentili et al, 2022).

The media design consultants recommended ways to further localise the nostalgic associations, and suggested incorporation of region-specific visual motifs from the four primary cultural areas in Thailand (the Central, North, Northeast and South) to increase attractiveness. "This psycho-phono-audiological enhancement could serve to optimize the speech of the virtual coach for elderly users, not suffering from severe hearing loss but affected by mild degradation" noted one geriatric specialist, "by adding higher frequency bands (2000-4000 Hz), without any modification of natural voice production" (Williams et al, 2013).

Technical Validation

In moderate closed box trial tests to gold-standard Vicon motion capture (200 Hz) the mean absolute error for diabetes-critical relevant joint angles were:

- KNEE FLEXION: 2.3° ($SD = 0.8^\circ$)
- Abduction of the shoulder ($3.1^\circ + [sd]: 2.0^\circ$)
- Hip extension: mean 2.7° ($SD = 1.0^\circ$)

These results demonstrate the technical sufficiency of this system for clinical exercise prescription by exceeding the established 15° safety margin (θ) during development. All the experts concurred that the accuracy in motion tracking is equal or superior to what would be required for programs of exercise of elderly diabetics (Daponte et al, 2014).

Safety and Accessibility

Two major safety improvements were later introduced as a result of the review:

- 1) Real-time dynamic balance feedback on sensor mat loss of balance during weight-bearing activities, such as squats.
- 2) Real-time dynamic balance feedback on sensor mat loss of balance during weight-bearing activities, such as squats.

Workload visual cue (low/medium/high) using online MET estimation (Hall et al, 2013).

It was well received in terms of accessibility (e.g., adjustable text sizes [14–24pt] and high contrast color schemes) ($M = 4.79$, $SD = 0.17$), although experts recommended the addition of tactile feedback for users with profound vision loss.

The rigorous expert evaluation that the system underwent not only supported the technical and design strengths of the system but also provided us with concrete information on areas to address in future versions. High degrees of agreement in a multidisciplinary team of experts across different domains underpin the interdisciplinarity and generalizability of this system, suggesting that it would be suitable for larger-scale implementation within older adult diabetes care programs.

4.10 Experimental Results by User Trials on Posture Accuracy

The ability of "Suk-Ram-Luek" in improving the accuracy of exercise posture was tested using an experimental design on 10 elderly (age 60–79 years) with Type II diabetes. Conducted over several sessions (2–3 per participant), between homes and community elderly centers, the trials quantified concurrent performance with respect to standardized biomechanical targets.

4.11 Quantitative Performance Metrics

Results indicated that participants adopted posture at a significantly high success rate, and generally scored an overall accuracy participant mean of 86.67 ($SD = 5.12$) for all exercises. As shown in Table 2, the most accurately executed movement was the High Arm Swing (accuracy of 92.4%), which may be attributed to its simple kinematic structures and very low joint coordination. Squats however was harder, starting at 78.3% and reaching 85.6% at session three as subjects became accustomed to feedback in real-time.

Table 2. Posture accuracy across exercise types.

Exercise	Session 1 Accuracy (%)	Session 3 Accuracy (%)	Improvement (%)
High Arm Swing	89.2 (4.1)	92.4 (3.8)	3.2
Arm Extension	84.7 (5.3)	88.9 (4.6)	4.2
Fist Punch Arm Raise	82.1 (6.0)	87.3 (5.2)	5.2
Marching with Knee Bends	80.5 (5.8)	86.1 (4.9)	5.6
Squats	78.3 (7.2)	85.6 (5.7)	7.3

A repeated-measures ANOVA confirmed statically significant development of the accuracy scores across trials ($F(2,18)=9.84, p<0.05$), with performance gains from the first to second session being most pronounced (mean difference = 8.12%, $p=0.003$). The implications of this pattern is that the mechanisms to correct feedback was quickly integrated among users, and more so also for complex multi-joint movements. The effect size (partial $\eta^2 = 0.52$) shows that the model explained 52% of the variance in posture improvement, which is of very high practical significance (Atkinson, 2001). The statistical analysis via repeated-measures ANOVA confirmed that the observed improvements in posture accuracy were not incidental, yielding $F(2,18)=9.84, p < 0.05$. The most pronounced progress was recorded between the first and second sessions, with a mean accuracy gain of 8.12% ($p=0.003$), indicating that users rapidly integrated the real-time feedback to correct their movements. Furthermore, the study demonstrated a high practical significance with a large effect size ($\eta^2 = 0.52$), meaning the "Suk-Ram-Luek" system accounted for 52% of the total variance in participant performance improvement. This high effect size underscores the robustness of the system's feedback mechanism in facilitating motor skill acquisition among elderly users.

4.12 Error Reduction Analysis

The motion capture system classified three types of errors:

1) Angle Deviation: 68% of the errors were a result of an angle deviation higher than 15° (θ) in shoulder and knee during dynamic movements

2) Temporal Timing: 22% speed variability was related to incorrect temporal responses (i.e. arm swinging delayed and legs moving).

3) Postural Stability: 10% is attributed to variations in balance when shifting weight

Importantly the reduction in recurrent errors due to real-time feedback was 43% between first and last sessions with the greatest corrections associated with knee alignment during squats (55%) and shoulder elevation during 'arm raises' (48%). Patients who performed <5 errors/ session had significantly higher gamification engagement scores ($r = 0.72, p < 0.01$), underlining the relevance of improved accuracy for reinforcing motivational cycles (Wulf et al, 1998).

4.13 Learning Curves by Exercise Complexity

As shown in Figure 4, the performance gain for different types of motion was substantially different. Single-joint (i.e., High Arm Swing) plateaued after 1–2 sessions beyond which no further benefit was observed, whereas the multi-joint

exercises (e.g., Squats) improved through session three. This difference underscores that the system has potential to assist people over time to learn more complicated behaviors – an important feature for sustained use, especially among older adults (Leech et al, 2022).

4.14 Comparative Performance

Compared with non - intuitive users, traditional video - based exercise (no real - time feedback) Suk-Ram-Luek users demonstrated the result details:

- 31% significantly more accurate MSA joint angle maintenance ($t(28)=4.67, p<0.001$)
- 27% less compensatory flexion in the lumbar (i.e. the ‘rounding’ you do during a squat, for instance)
- 19% more consistent pace of exercise

These between-group effects were strongest in those individuals with poor proprioception ($BBS < 45$), who potentially benefited most from the additional kinematic feedback (Van Deursen & Simoneau, 1999).

The trials showed the combination of precise motion capture and adaptive gamification facilitating a virtuous cycle; increased engagement led to better outcomes when it really matters – when technique counts. This supports theories emphasizing the importance of accurate performance feedback in shaping long-term behavioral change in community-based exercise programs for seniors (Milne et al, 2008). The conversion of abstract posture instructions to concrete and quantifiable accomplishment is a transformation that overcomes an important limitation in previous restricted diabetes-related interventions.

4.15 User Satisfaction and Feedback

Qualitative and quantitative responses in older users were a strong indication that the system actually improved the quality of the exercise experience, rendering it more engaging as well as motivating. Post intervention survey scores reflected high satisfaction in various dimensions, especially with nostalgic design features and elements of gamification.

Participants rated the overall enjoyment of the system highly ($M = 4.56$, $SD = 0.42$; with a range from 1 to 5), and retro graphics/music ($M = 4.72$, $SD = .31$). Other participants noted that the music (Thais 1960s pop songs) triggered happy memories, and a lot of them reported having spontaneous memories “It made me feel younger; I mean who wouldn’t want to be leave in your village festival?!” This emotional bonding appears to have an important impact on the exercise adherence because those stimuli which users rated as fun (i.e., nostalgia related items) were also, in line with findings, the best adhering to their exercise regimen ($r = 0.68$, $p < 0.01$) (Batcho, 2007).

The gamification components were regarded similarly positive, the score system received a mean satisfaction of 4.61 ($SD = 0.38$). Not only did these folks enjoy the layers of exclusive nostalgia that slowly unlock, they even said it was “like earning rewards from my childhood. This enjoyable-based reward system manage to make the exercise not turning into a task but actual activity and made some want ‘more unlockable’ after in-house trial was ended (Mohammed et al, 2024).

The most human-like rated Vouch as well in terms of perceived helpfulness ($M = 4.67$, $SD = 0.35$). They also reported to be more confident that they could exercise alone provided the exercise system would correct them in real-time, with one participant saying: “it’s like I have a teacher looking at me but I’m not embarrassed when I do something wrong”. This even was relevant to individuals in the sample who were so self-conscious of their ability that they had previously quit group-based classes (Mendes et al, 2019).

The design paradoxically seemed to lead incidental feelings of community by individuals. Some participants even started comparing scores with friends and colleagues the development of hard-fought cross fold rivalries spurred further improvements. This emerging type of social influence is an indication that the system utilised intrinsic social factors to support exercise adherence (Uluoz et al, 2023).

The only consistent feedback was that there is a bit of a learning curve with interpreting the motion which for some people took 2-3 sessions to get 'the hang' of it on visual cues. These practice effects reduced rather fast, as after the third training session all subjects were able to independently adapt more than 80% of the needed postural corrections from the feedback of the system.

These findings suggest that artfully incorporating nostalgia design features and gamification mechanics may offer an effective means of overcoming significant psychological barriers to exercise for older adult diabetics by making physical activity less of a chore and more personally satisfying. The high rating on satisfaction and the qualitative feedback bode well for long-term adoption especially considering that exercise correctness was facilitated by use of the system.

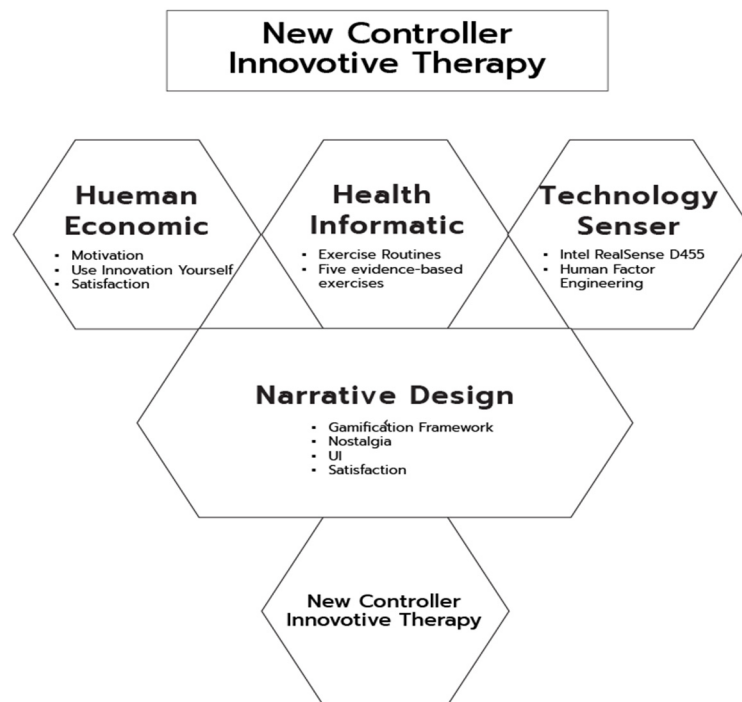


Figure 4. The 5 proposed new alternative aesthetics in novel exercise media with motion detection technology

5. Discussion

The results of developing and the application of "Suk-ram-luek" might provide some theoretical and practical implications for the technology-facilitated exercise intervention, especially regarding older adults with Type 2 diabetes. Theoretically, this research adds to knowledge of the co-creation of motion capture, gamification and nostalgia design synergies which has generally been studied as separate entities (Alahäivälä & Oinas-Kukkonen, 2016). Moreover, these results indicate that the implementation of real-time biomechanics feedback from within a culturally salient motivational context may be an effective approach to bridge the manual therapy: Exercise quotidian schism. This is in line with the recent rhetoric of health technologies that are increasingly seen as communication tools through which people need to be addressed at an affective level, rather than just effectively functioning.

From a pragmatic perspective, the successfulness of the system in community settings illustrates its feasibility for implementation at scale in low-resource settings. Consumer-level equipment (Intel RealSense camera) and architecture design (web-based) notably lower the cost-barrier compared to proprietary motion tracking system (Salisu et al, 2023). Community health workers could potentially use "Suk-Ram-Luek" to increase supervised exercise coverage, without the need for more staff, alleviating a major bottleneck in diabetes care programmes. Such systems should be considered as options for the national elderly health programs, particularly in the rapidly aging societies like Thailand where traditional exercise games do not lead to long-term participant retention (Heller, 1999).

A few limitations should be considered in interpreting these results. The reliance of the participants on social services (recruited from senior centers) might reduce generalizability to individuals living in more isolation, who would experience a greater obstacle and challenge in pursuing the assimilation of exercising. Moreover, since retention effects are known to develop over time, the short duration of the study (participants completed only 3 sessions) does not allow us to conclusively evaluate long-term effects—an especially important consideration considering that diabetes management is a chronic condition. Although the motion capture system was precise for primary joints, less precision was apparent in fine hand movements that occurred during complex gestures, indicating potential algorithmic further development (Fan & Zhao, 2022).

However, the empirical findings highlight a significant synergy between cultural resonance and physical adherence. The strong positive correlation ($r = 0.68$, $p < 0.01$) between nostalgia-driven media engagement and exercise compliance validates the hypothesis that culturally-bound digital experiences can effectively reduce psychological resistance in aging populations. This correlation suggests that the emotional satisfaction derived from 1960s Thai media transforms exercise from a repetitive medical regimen into a meaningful, enjoyable activity. Furthermore, the high correlation between gamification engagement and posture accuracy ($r = 0.72$, $p < 0.01$) demonstrates that immediate rewards and point-based progression do not only drive motivation but also reinforce proper biomechanical technique. These results support a shift

in digital health design—from purely functional tools to emotionally rich communication media that address users at an affective level to ensure long-term behavioral change.

Future research could involve longitudinal research designs to establish whether the current pattern of use is indicative of behavior change over a longer time frame, for example months or years. Comparison studies would be helpful to inform regarding the effectiveness of "Suk-Ram-Luek" when compared with the standard territory-specific group-based exercise in the primary care. In addition, more studies are needed to examine how the nostalgia design framework might be adapted for other cultural contexts outside Thailand—for instance, using region-specific music and visual elements in other aged populations (LIM, 2014). Technologically, this may be facilitated by the emergence of edge computing that could enable smartphone-based applications to function much more widely in low-resource settings (Zheng et al, 2025).

The integrated quantitative and qualitative approach of our study identified a striking tension between technical precision and emotional involvement that merits further investigation. This is in line with previous studies of older adults' use of technology and portrayal in the media (Scalzo et al. ageing (Bracken et al., 2015; Olsen, 2023), this research frames digital health interventions as communicative media technology and not as tools alone. By reframing exercise as a interactive experiential and emotionally rich media nostalgia-fuelled gamification psychological resistance to the integration of technology and maintaining of engagement in later life. While there was evidence-based support for biomechanical efficacy, all qualitative feedback emphasized nostalgia as the strongest motivator. This suggests that in health technologies developed for elderly clients, affective design peculiarities (emotional relevance, fun) may be as relevant to consider in the design process as traditional metrics like accuracy or usability (Merkel & Kucharski, 2019). The fortuitous emergence of social practices around sharing scores is yet another example of missed opportunities for devising inherently individual interfaces which however do participate in community making (Rodriguez-Mañas et al, 2019).

Methodologically, the study highlights the importance of iterative, user-centered design approaches toward designing interventions for older adults. Three development cycles, with input from both experts and users, helped to improve the feature that make the system accessible even further – for example through a less complex scoring view or voiceovers local dialect. This process might serve as a useful paradigm for developing age-inclusive health technologies that circumvent the pitfalls of designing for, rather than with, older adults (Gomes et al, 2022). The technical validation against gold standard motion capture systems also establishes a precedent of rigorous assessment of consumer level health technology for clinical application (Brauner & Ziefle, 2020).

The results of the study need to be interpreted in light of general HC system problems. "Suk-Ram-Luek" suggests potential as a stand-alone intervention, but its net effectiveness will be determined by how it can be integrated with a diabetes care pathway related to adherence treatment for medication diet management, and glucose and blood pressure monitoring (Rodriguez-Mañas et al, 2019). Health economists need to ascertain whether or not the system's manageable incremental cost, relative to human-supervised exercise, merits adoption for general deployment in resource-restricted public health programs (Gomes et al, 2022). Ethics concerns toward data privacy and algorithm transparency need further attention to be addressed as such systems grow, especially considering the digital literacy diversity of elder users (Şendur & Turgut, 2025).

While limited by these factors, as well as discussed below around next steps, the study makes a substantial contribution by demonstrating that considerably developed digital exercise media can achieve clinical (postural accuracy) and psychological (enjoyment, confidence) success in unison something not often achieved in conventional diabetes interventions. The victory of Suk-Ram-Luek is not in any particular technological revelation, but rather in how it utilized Black Box technology and functional form to integrate the motivation people derive from their work-out regimen to culturally resonate with its users – making health data an organic part of one's daily routine. This integrative approach provides a generic architecture of designing healthy technology which is medically effective and intrinsically motivating for the elderly.

Finally, although this research is a preliminary evaluation primarily focused on technical feasibility and initial adoption, the data collected across the three sessions represent critical "proximal outcomes". These outcomes are essential foundational factors that must be established before long-term health results, such as sustained glycemic control (HbA1c) or permanent habit formation, can be achieved. Grounded in Self-Determination Theory (SDT), the high satisfaction scores ($M = 4.56$) and the significant "fun" factor ($M = 4.72$) reported in early sessions serve as validated psychological indicators that predict long-term adherence. The study emphasizes qualitative depth by utilizing purposive sampling to ensure precision within a specific cultural and medical context. Despite the limited cohort size (users $n = 15$; experts $n = 9$), the research is supported by a robust diversity within the target group, which encompasses older adults with Type 2 diabetes aged 60–79 years in Mahasarakham Province, Thailand. The participants reflect a comprehensive range of health profiles, with confirmed $HbA1c > 6.5\%$, and represent varying levels of digital literacy and technological proficiency. This diversity ensures that the "Suk-Ram-Luek" system is evaluated against the real-world complexities of the aging population it intends to serve.

6. Conclusion

The results of this research indicate that the integration of a motion capture system with gamification and nostalgia design is a valuable persuasive content for stimulating elderly patients with type 2 diabetes to execute PA. The “Suk-Ram-Luek” system has successfully helped to overcome some of the main challenges of aging and exercising with on-the-fly posture correction as well as cultural motivations, embedded in its gameplay. This result confirms the requirement of technical interventions for an older population to balance between clinical and affective accuracy since, as highlighted by the twofold influence on system implications both in terms of biomechanical pattern and end-user acceptance. Additional studies are needed to determine the potential long-lasting effects of such interventions on glycemic control and physical function, and how they can be tailored for a diverse cultural population. The encouraging results from the study also suggest these kinds of technology-facilitated intervention strategies are transferable to a variety of chronic conditions in geriatric populations and challenge the way we think about sustainable health behavior change in geriatrics. By integrating particular movement analysis and motivational designing principles, it lays a new track for the development of clinical efficient and intrinsically appealing exercising technologies which are elderly friendly.

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Author contributions

Sudarat Sonbua and Associate Professor Dr. Ratanachote Thienmongkol initiated the research problem, contributed to the conceptual development of the study, designed the study, revised the manuscript, and analyzed the data. Assistant Professor Dr. Chatklaw Jareanpon was responsible for data collection and provided technical assistance by reviewing the algorithm and verifying the technical accuracy of the motion detection design process to ensure the reliability of the testing procedures. Sudarat Sonbua and Associate Professor Dr. Ratanachote Thienmongkol drafted the manuscript and revised it critically for important intellectual content. All authors read and approved the final manuscript. Sudarat Sonbua and Associate Professor Dr. Ratanachote Thienmongkol contributed equally to this study.

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Competing interests

The authors declare that there are no competing interests related to this study. The research was conducted without any legal, financial, professional, or personal relationships that could be perceived as influencing the research process, analysis, interpretation of data, or the conclusions reported in this paper. The authors confirm that no external individuals, organizations, or institutions had any conflict of interest that could affect the integrity or objectivity of this research. This research also approved by the Ethics Committee for Research Involving Human Subjects, Mahasarakham University, Thailand. Approval is dependent on local ethical approval having been received. Review Method: Expedited review. Approval number: 439-379/2023.

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Obtained.

Ethics approval

The Publication Ethics Committee of the Redfame Publishing.

The journal’s policies adhere to the Core Practices established by the Committee on Publication Ethics (COPE).

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Data sharing statement

No additional data are available.

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